

BASIC NETWORK SCANNING EXERCISE

1. Introduction

Network scanning is a basic cybersecurity technique used to identify active hosts, open ports, and network services on a computer or network.

In this exercise, I used **Nmap (Network Mapper)** to scan my own local computer. The target used was 127.0.0.1, which represents the localhost of the computer.

The purpose of the exercise was to understand how open ports and network services can be identified and why unnecessary services should be properly secured.

2. Objectives

The main objectives of this exercise were:

- To understand the concept of network scanning.
 - To learn how to use Nmap.
 - To identify open TCP ports on a local computer.
 - To identify services associated with open ports.
 - To understand possible security risks.
 - To document the scan results.
 - To learn basic security recommendations for exposed services.
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3. Tool and Environment

Tool

Nmap (Network Mapper)

Nmap is a network scanning tool used for network discovery and service identification.

Environment

- Operating System: Windows
- Nmap Version: 7.991
- Target: 127.0.0.1
- Target Type: Local computer

The scan was performed only on my own computer.

4. Methodology

First, I verified that Nmap was installed correctly.

The following command was used:

```
nmap --version
```

After confirming the installation, I performed a basic scan of the local computer:

```
nmap 127.0.0.1
```

I then performed service and version detection:

```
nmap -sV 127.0.0.1
```

The -sV option allows Nmap to attempt to identify the services and versions running on open ports.

5. Scan Results

The Nmap scan identified one active host and five open TCP ports.

Port	Protocol	State	Nmap Detection
135	TCP	Open	Microsoft Windows RPC
445	TCP	Open	Microsoft-DS?
1042	TCP	Open	afrog?
1043	TCP	Open	ssl/boinc?
7778	TCP	Open	interwise?

The scan also reported that **995 TCP ports were closed with connection refused.**

6. Analysis of Findings

Port 135 – Microsoft Windows RPC

Port 135 is commonly associated with Microsoft RPC services. RPC allows Windows components and services to communicate with each other.

Because this is a Windows system service, it should be properly protected by firewall and network configuration.

Port 445 – Microsoft-DS

Port 445 is commonly associated with SMB (Server Message Block), which is used by Windows for functions such as file and printer sharing.

SMB services should be appropriately secured and restricted to trusted networks when possible.

Port 1042

Nmap associated this port with afrog, but the question mark indicates that the service could not be confidently identified.

During service detection, the port returned an HTTP response. Therefore, it appears that some type of web service is listening on this port.

The exact application was not confirmed by Nmap.

Port 1043

Nmap associated this port with ssl/boinc?, but the question mark indicates that the identification was uncertain.

The scan received an HTTP response over an SSL/TLS connection, suggesting that a web-based service may be listening on this port.

The exact application was not confirmed.

Port 7778

Nmap associated this port with interwise?, but the service could not be confidently identified.

Further investigation would be required to determine which application is using this port.

7. Potential Security Risks

An open port does not automatically mean that a system is vulnerable. The actual risk depends on the service, software version, configuration, firewall rules, and network accessibility.

Potential risks identified during the exercise include:

- Unnecessary open ports can increase the system's attack surface.
- Outdated services may contain security vulnerabilities.
- File-sharing services such as SMB should be properly configured.
- Unknown services should be investigated to determine whether they are necessary.
- Services exposed to untrusted networks can create additional security risks.

8. Security Recommendations

The following security practices are recommended:

1. Keep Windows and installed applications updated.
2. Use Windows Firewall and review its rules.
3. Disable unnecessary network services.
4. Restrict file and printer sharing to trusted networks.
5. Investigate unfamiliar open ports.
6. Regularly review applications that listen for network connections.
7. Avoid exposing unnecessary services to untrusted networks.

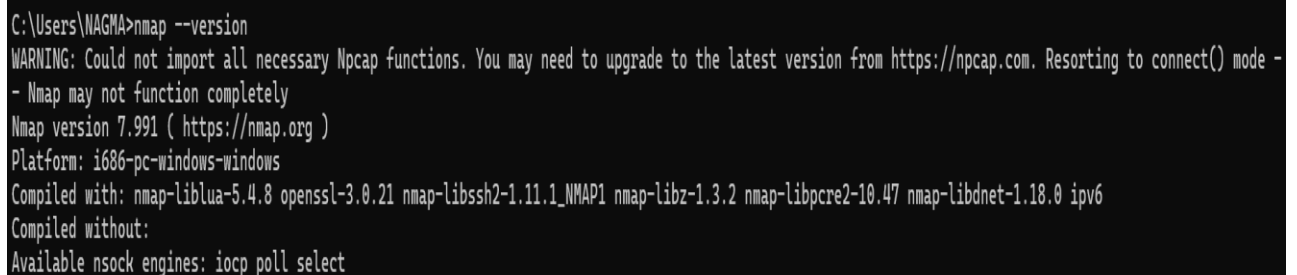
9. Screenshots / Evidence

Figure 1: Nmap Version

The screenshot shows that Nmap 7.991 was installed successfully.

Command:

nmap --version



```
C:\Users\NAGMA>nmap --version
WARNING: Could not import all necessary Npcap functions. You may need to upgrade to the latest version from https://npcap.com. Resorting to connect() mode -
- Nmap may not function completely
Nmap version 7.991 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.21 nmap-libssh2-1.11.1_NMAP1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select
```

Figure 2: Basic Nmap Scan

The screenshot shows the basic scan performed against the local computer.

Command:

nmap 127.0.0.1

```
C:\Users\NAGMA>nmap 127.0.0.1
WARNING: Could not import all necessary Npcap functions. You may need to upgrade to the latest version from https://npcap.com. Resorting to connect() mode
- Nmap may not function completely
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-08 17:23 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00083s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
1042/tcp   open  afrog
1043/tcp   open  boinc
7778/tcp   open  interwise
```

Figure 3: Service and Version Detection

The screenshot shows the services detected by Nmap on the open ports.

Command:

nmap -sV 127.0.0.1

```
Command Prompt
C:\Users\NAGMA>nmap -sV 127.0.0.1
WARNING: Could not import all necessary Npcap functions. You may need to upgrade to the latest version from https://npcap.com. Resorting to connect() mode -
- Nmap may not function completely
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-08 17:47 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00073s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
445/tcp    open  microsoft-ds
1042/tcp   open  afrog?
1043/tcp   open  ssl/boinc?
7778/tcp   open  interwise?
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port1042-TCP:V=7.991I=7%D=9/8%Time=6A9FFD01P=i686-pc-windows-windows%
SF:p(GetRequest,2AE,"HTTP/1.1\x20404\x20Not\x20Found\r\nVary:\x20Origin\r
SF:\nContent-Security-Policy:\x20default-src\x20'none'\r\nCross-Origin-Res
SF:ource-Policy:\x20cross-origin\r\nOrigin-Agent-Cluster:\x20\?1\r\nReferr
SF:er-Policy:\x20no-referrer\r\nStrict-Transport-Security:\x20max-age=1555
SF:2000;\x20includeSubDomains\r\nX-Content-Type-Options:\x20nosniff\r\nX-D
SF:NS-Prefetch-Control:\x20off\r\nX-Download-Options:\x20noopen\r\nX-Frame
SF:Options:\x20SAMEORIGIN\r\nX-Permitted-Cross-Domain-Policies:\x20none\r
SF:\nX-XSS-Protection:\x200\r\nContent-Type:\x20text/html;\x20charset=utf-
SF:8\r\nContent-Length:\x20139\r\nDate:\x20Tue,\x2008Sep\x202026\x2012
SF:18:09:18GMT\r\nConnection:\x20close\r\n\r\n<!DOCTYPE\x20html>\n<html>
SF:\x20lang=en">\n<head>\n<meta\x20charset=utf-8">\n<title>Error</tit
SF:le>\n</head>\n<body>\n<pre>Cannot\x20GET\x20/</pre>\n</body>\n</html>\n
SF:")&r(HTTPOptions,D2,"HTTP/1.1\x20204\x20No\x20Content\r\nVary:\x20orig
SF:in,\x20Access-Control-Request-Headers\r\nAccess-Control-Allow-Methods:\
SF:\x20GET,HEAD,PUT,PATCH,POST,DELETE\r\nContent-Length:\x200\r\nDate:\x20T
SF:ue,\x2008Sep\x202026\x2012:18:09:18GMT\r\nConnection:\x20close\r\n
SF:ri:n")&r(RTSPPRequest,D2,"HTTP/1.1\x20204\x20No\x20Content\r\nVary:\x20
SF:Origin,\x20Access-Control-Request-Headers\r\nAccess-Control-Allow-Metho
SF:ds:\x20GET,HEAD,PUT,PATCH,POST,DELETE\r\nContent-Length:\x200\r\nDate:\
SF:\x20Tue,\x2008Sep\x202026\x2012:18:09:18GMT\r\nConnection:\x20close
SF:ri:n\r\n")&r(RPCCheck,2F,"HTTP/1.1\x20400\x20Bad\x20Request\r\nConnect
SF:ion:\x20close\r\n\r\n")&r(DNSVersionBindReqTCP,2F,"HTTP/1.1\x20400\x20
SF:Bad\x20Request\r\nConnection:\x20close\r\n\r\n")&r(DNSStatusRequestTCP,
```

10. Conclusion

In this exercise, I learned how to perform basic network scanning using Nmap. I scanned my own local computer using the localhost address 127.0.0.1.

The scan identified five open TCP ports: 135, 445, 1042, 1043, and 7778. I learned how Nmap can identify open ports and attempt to determine the services running on them.

I also learned that an open port does not automatically represent a vulnerability. Each service should be investigated and properly configured according to its purpose.

This exercise provided practical knowledge about network discovery, ports, services, and basic network security.

11. Ethical Consideration

This exercise was performed only on my own local computer.

Network scanning should only be performed on systems or networks that you own or have explicit permission to test.